Probing the Earth with Nuclear Explosions[1]

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Abstract. Progress in seismology is reviewed with emphasis on the usefulness of past nuclear weapon tests in determining the internal structure of the earth. Shot times and locations are tabulated for 169 U. S. atomic explosions, with seismic data from Pasadena. The advantages of using large chemical explosions and future nuclear explosions detonated under the Plowshare program as controlled energy sources for carefully instrumented seismological experiments are discussed. Finally, an international program of explosions for seismological research is proposed, and specific suggestions are made for attacking several outstanding problems in seismology by means of chemical and clean nuclear explosions and the instrumentation network proposed at Geneva for nuclear test detection. Deep oceanic seismometer lines are proposed as a means for making important improvements in seismic knowledge of the world and possible improvements in the detection of atomic explosions.

INTRODUCTION

Our knowledge of the constitution of the earth has been pieced together from the evidence afforded by various branches of science. By far the most detailed information, however, has come from seismology, which can in principle provide detailed information about every element of volume of the earth commensurate with the wavelength transmitted. Prior to the atomic age, earthquake seismology had proved that the earth consisted of a crust, a mantle, and an outer and inner core, and had provided much information as to the variation of seismic velocities within each shell.

Bullen [1956], making use of the *Adams and Williamson* [1923] relation between velocities and density, calculated the density distribution in the earth consistent with these seismic velocities, the moment of inertia, and plausible assumptions as to the composition of each shell.

Birch has led in the laboratory determination of elastic constants at elevated pressures and temperatures. He has also [*Birch*, 1952] led in applying theory of finite strain to the seismic data in order to extract still more physical information. He concludes:

1. The mantle is probably homogeneous from 900- to 2900-km depth.
2. The mantle above 900-km depth is not homogeneous, and must vary in composition or in phase, or both.

The seismic data of the pre-atomic age were not sufficiently accurate to establish this deduction beyond doubt, nor was the laboratory knowledge of equation of state adequate to test the theory.

The importance of knowing whether or not the mantle is homogeneous lies in the fact that all theories of the origin and development of the earth, of the origin of continents and oceans, and of mountain building depend vitally on this knowledge.

As an example, *Rubey* [1951] advanced a compelling argument that the ocean has grown at a nearly constant rate throughout geologic time by abstraction of water from the mantle. This volcanic water is presumed to have been removed from the mantle by a process of selective fusion and recrystallization. Such a process would also segregate the lower melting silicate material at a proportionate rate, and this is presumed to have formed the continental crust.

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Such abstraction of material from the mantle must either result in different composition under the continents than under the oceans, or there must be motion in the mantle, as by convection, to remove such differences. The uncertainties of seismic velocities obtained from natural earthquakes were greater than the expected differences. (Recently, mantle Rayleigh waves were used to establish a difference in the outer mantle under continents and under oceans [*Dorman, Ewing, and Oliver*, 1960].)

Let us now look at the beginning that has been made in atomic seismology and then consider what might be done in the future. In this discussion we naturally presuppose that nuclear explosions for peaceful purposes will be conducted under conditions such that no radiation hazards will occur.

Seismic Results to Date from Atomic Explosions

Bullen [1958], in his presidential address to the International Association of Seismology and Physics of the Interior of the Earth, recognized the tremendous role atomic explosions could play in geophysical research:

> In seismology, where the release of energy at a natural earthquake focus is unseen and uncontrolled, we battle hard to reduce uncertainties in our inferences from the data Nature provides us. In studies of the Earth's outer layers, we have contrived some experimental control at source by using limited artificial explosions to complement and supplement our findings from natural earthquakes. Since 1945, we have glimpsed the possibility of extending this control to the point where waves of seismic type can be sent artificially to the very deep interior and be recorded at the surface to tell their tale. . . . . Already we have glimpsed the possibility of controlled experiments which can give us the increased resolving power that we need so badly in problems of the deeper interior.
>
> We have had even more than a glimpse. Some nuclear explosions have already yielded results of value to seismology, even though the seismological applications to date have come only as the merest incidentals to the central purposes of the explosions.

The first atomic explosion, Trinity, near Alamagordo, New Mexico, July 16, 1945, was seismically recorded. It provided some corroboration of deductions from earthquakes, but no important new knowledge.

Bikini Baker, the underwater explosion of July 24, 1946, was the first atomic explosion for which location and origin time were released. This 20-kiloton explosion gave P wave amplitudes at large distances equal to those from an earthquake of magnitude 5.5 [*Gutenberg and Richter*, 1946]. The energy-magnitude relation then used gave the elastic-wave energy of such an earthquake as 8×10^{21} ergs, or 200-kt equivalent energy, whereas we know that Bikini Baker coupled only about 0.1 kt of energy to the elastic-wave field. This was one of the strong stimuli toward the downward revision of the earthquake energy scale, which has since been reduced by a factor of nearly a thousand.

Bikini Baker also showed negative residuals of 1.8 ± 0.8 sec against the Jeffreys-Bullen travel-time tables, at distances of 70° to 80° (7700–8800 km). In other words, the P waves arrived 1.8 seconds sooner than was expected for an 'average' earthquake.

Gutenberg [1952] reported some seismic results for five airburst shots of the Ranger series at the Nevada test site, early in 1951. The results indicated a crustal thickness of 35 to 40 km and a sub-Moho P-wave velocity of 8.2 km/sec—the first explosion sounding of the crust in this region.

The first hydrogen bomb test, Ivy-Mike at Eniwetok Atoll on October 31, 1952, was also the first artificial explosion recorded the world over. No seismic results were published, probably for ethical reasons.

When the Castle series of hydrogen bomb tests was announced by the United States in 1954, *Burke-Gaffney and Bullen* [1957] were able to deduce the origin time to 0.3 sec, and hence they were able to use the world-wide recordings for seismological purposes. They confirmed the negative residuals found from Bikini Baker and made necessary a revision of the travel-time tables for surface epicenters in the mid-Pacific. Second, the travel time from Bikini to Australia and to the United States is the same to 0.5 sec, indicating that the average velocities are the same to 0.1 per cent. When the actual times of the detonations were released, they agreed with the deduced times to within 0.3 sec.

Most interesting of all, stations at Pretoria,

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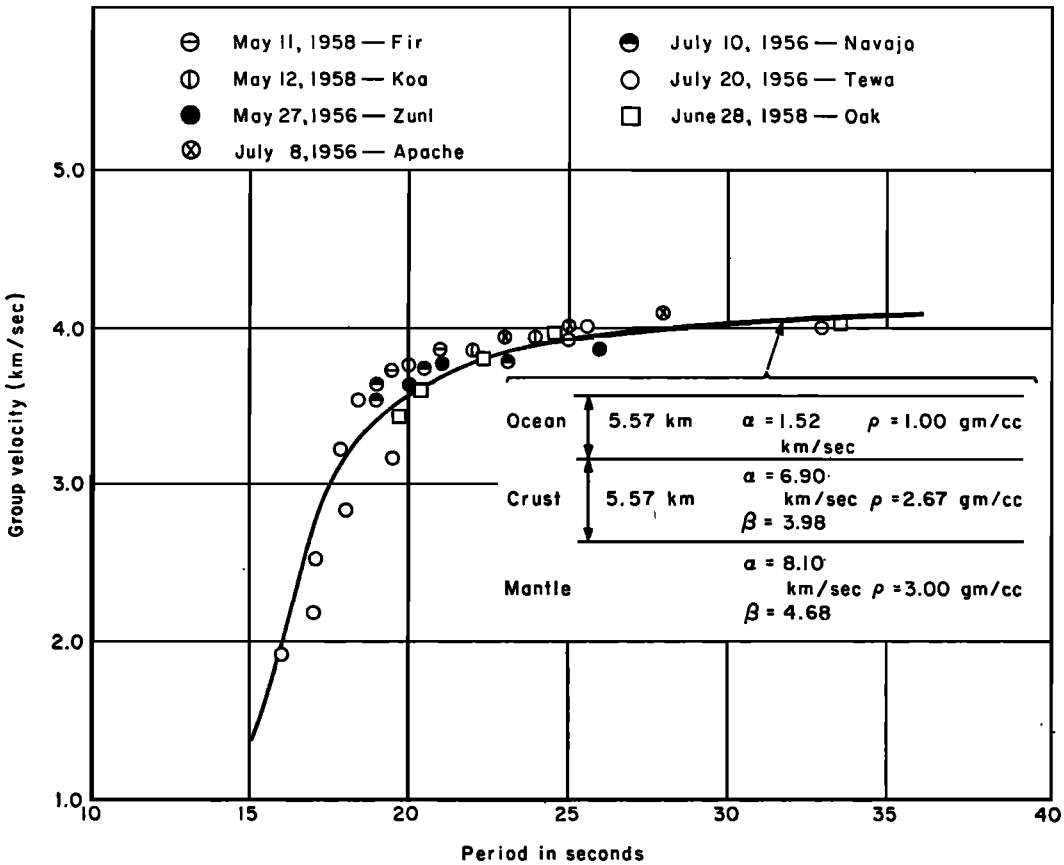

Fig. 1. Surface wave dispersion from Pacific shots recorded at Pasadena.

Kimberly, and Tamanrasset, at distances of 137° to 142°, all showed precursors about 11 sec ahead of the scheduled *P*-wave arrival times and 3 sec ahead of *PKIKP*. These were shown to be *PKP* waves diffracted from the outer core. It had been suggested that the waves thought to be propagated through the inner core (*PKIKP*) might be diffracted *PKP* waves. Hence the identification of these two separated sets of waves added the 'final link in the chain of evidence for the existence of the earth's inner core' [*Burke-Gaffney and Bullen*, [1958].

As the Atomic Energy Commission began declassifying the times and locations of test explosions in the Pacific and Nevada proving grounds, seismological articles began to appear based on records obtained in the routine earthquake watch. *Oliver and Ewing* [1958*a*] observed surface waves at Palisades, New York, from the Wigwam underwater explosion off the coast of California. This source excited two types of surface waves which had not previously been found to propagate along oceanic paths—the short-period branch of the Rayleigh wave and the first shear mode. This knowledge aids in the interpretation of the dispersion of earthquakes and in the use of the dispersion method to determine properties of the crust and mantle.

To illustrate the possibilities of nuclear explosions, we present in Figure 1 the Rayleigh mode dispersion for the oceanic paths between the test sites and Pasadena for six events. The theoretical dispersion curve for the case shown below the curve represents a very good fit to the data. These results place on a firm basis the corresponding results from earthquakes and make possible a precision heretofore not available. Small deviations for different regions must

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(in the case of artificial sources) represent effects of propagation rather than uncertainties in epicenter or origin time. The interpretation of these deviations will yield information on regional differences in crustal and mantle structure.

Furthermore, the corresponding seismograms represent the impulse response of the wave guide, in this case the oceanic crust and mantle. In principle, the impulse response can be used to separate the effects of propagation from the effects of the source. Thus, surface waves can be used to deduce the mechanism of the source, and they may eventually provide more information than body waves.

Oliver and Ewing [1958*b*] also discussed surface waves from explosions in the Marshall Islands and at the Nevada test site. *Carder and Bailey* [1958] further refined the travel-time curves with data from explosions. The British and Australian governments cooperated in the seismic exploitation of four atomic tests at Maralinga in central Australia in 1956. Appropriate instruments were deployed in such a way as to give maximum information on continental and outer-mantle structure. The result is the first really reliable knowledge of deep continental structure and of mantle velocity in the whole Australian continent [*Doyle*, 1957].

Bullen [1958] summarized the importance of this as follows: 'In Australia, we are, by world standards, very free from earthquakes, and we are also, as yet, short of having an adequate network of observatories. . . . Thus the 1956 explosions provide one instance in which atom bomb explosions have been turned to great advantage, even though the results relate only to near-surface structures.'

With the announcement of the Rainier underground explosion and the Hardtack II series at the Nevada test site, many seismological studies were undertaken, and they have been or will be reported in the literature. Outstanding among these is the work reported by *Romney* [1959].

Romney's account of travel-time and amplitude determination across the United States from the Hardtack II shots will stand as a classic in seismology. Never before have such extensive and accurate records been obtained on a continental scale. Even so, they leave much to be desired, and very much more could be accomplished with a program of atomic explosions for seismic purposes, supported by adequate seismic instrumentation.

Future Possibilities

Nearly 300 atomic tests have been conducted so far by Russia, Britain, and the United States. In Table 1 169 tests performed by the United States are listed.[2] The time and location are given for each, together with some yields and all Pasadena seismic amplitude data. We believe that, because of the secrecy surrounding these tests, seismologists have not sought to glean all the possible information from their records of these tests. We hope that publication of these data, which contain much that has not hitherto been released, will stimulate seismologists to look back into their records to extract more information from these old shots.

Table 1 contains shots of all kinds—high-, medium-, and low-altitude, surface, underground, and underwater—and gives the available yield and Pasadena ground-motion amplitude for P and Rayleigh waves. These amplitudes are not those of the Fourier component for the period shown and must therefore be considered rough indications to be used for comparison only.

In Table 2 the seismic coupling for shots of one yield under various conditions are roughly compared. Application of the earthquake magnitude scale to explosions is highly questionable. One approach is to stipulate that the magnitude must correspond to that of a normal earthquake at the same distance which has excited P waves of the same amplitude and period. In the future it will be better to work directly with energy. The partition of energy between P, S, and surface waves will be different for earthquakes and explosions. It should be remarked here that all the underground shots have been in dry desert alluvium or in porous volcanic tuff. The energy coupled to seismic waves from these shots may not be typical of explosions in other types of rock. The Gnome experiment in a compact salt bed would provide more information on this point. For this and other reasons, Table 2 should be viewed as a crude comparison of explosions under varying source conditions.

In terms of distant P-wave amplitudes, a

[2] This table is based on shot information compiled by G. W. Johnson for presentation to the International Union of Geodesy and Geophysics at Helsinki, 1960.

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TABLE 1. Nuclear Explosions of Possible Interest to Seismologists Conducted by the United States

Shot Name	Date	Time, UT	Coordinates	Height of Burst, ft.	Type of Burst	Yield Mt	Yield kt	Yield tons	P wave* Amplitude, μ	P wave* Period, sec	Rayleigh Wave† Amplitude, μ	Arrival of Air Shock at Ground
Operation Trinity												
Trinity	7/16/45	12:29:15	33°40′31″N 106°28′29″W	100	Tower		19.3					
Operation Crossroads												
Able	6/30/46	21:01:	Bikini	520	Air drop		Nominal					
Baker	7/24/46	20:35:	Bikini	−90	Underwater		Nominal					
Operation Sandstone												
X-ray	4/14/48	18:16:59	Eniwetok	200	Tower		36.5					
Yoke	4/30/48	18:08:59	Eniwetok	200	Tower		48.7					
Zebra	5/14/48	18:04:00	Eniwetok	200	Tower		18.2					
Operation Ranger												
A	1/27/51	13:44:51.0	36°48′N 115°57′W	1,060	Air drop		1.27		0.15	0.7		
B_1	1/28/51	13:52:04.5	36°48′N 115°57′W	1,080	Air drop		7.83	0.72	0.72	0.7		
E	2/1/51	13:46:39.5	36°48′N 115°57′W	1,080	Air drop		1.00		0.15	0.7		
B_2	2/2/51	13:48:48.0	36°48′N 115°57′W	1,100	Air drop		7.95		0.51	0.6		
Freddie	2/6/51	13:46:55.0	36°48′N 115°57′W	1,435	Air drop		22.2		1.20	0.7		
Operation Greenhouse												
Dog	4/7/51	17:33:57.8	11°33′21″N 162°21′16″E	300	Tower							
Easy	4/20/51	17:27:00.1	11°40′08″N 162°14′25″E	300	Tower		46.7					
George	5/8/51	20:30:00.7	11°37′37″N 162°18′53″E	200	Tower							
Item	5/24/51	17:16:59.3	11°40′23″N 162°14′55″E	200	Tower							
Operation Buster-Jangle												
Able	10/22/51	14:00.	37°05′02″N 116°01′26″W	100	Tower		<0.1					
Baker	10/28/51	15:20:08.9	37°05′06″N 116°01′12″W	1,118	Air drop		3.5					

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TABLE 1. Continued

Shot Name	Date	Time, UT	Coordinates	Height of Burst, ft.	Type of Burst	Yield Mt	Yield kt	Yield Tons	P wave* Amplitude, μ	P wave* Period, sec	Rayleigh Wave† Amplitude, μ	Arrival of Air Shock at Ground
Charlie	10/30/51	15:00:29.8	37°05′06″N 116°01′13″W	1,132	Air drop		14.0		0.06	1.0		
Dog	11/1/51	15:30:01.6	37°05′05″N 116°01′11″W	1,417	Air drop		21		0.08	1.0		
Easy	11/5/51	16:29:58.2	37°05′31″N 116°01′28″W	1,314	Air drop		31.4		0.15	1.0		
Sugar	11/19/51	16:59:59.7	37°07′54″N 116°02′19″W	4	Surface		1.2					
Uncle	11/29/51	19:59:59.7	37°10′11″N 116°02′33″W	−17	Underground		1.2					
Operation Tumbler Snapper												
TS-1	4/1/52	17:00:07.5	36°47′54″N 115°56′08″W	793	Air drop		1.1		0.04	0.5		
TS-2	4/15/52	17:29:57.1	37°05′03″N 116°01′10″W	1,109	Air drop		1.2					
TS-3	4/22/52	17:30:10.0	37°05′04″N 116°01′13″W	3,447	Air drop		30.7		0.03	0.7		
TS-4	5/1/52	16:29:59.1	37°05′03″N 116°01′13″W	1,040	Air drop		19.2		0.04	0.7		
TS-5	5/7/52	12:14:59.3	37°03′11″N 116°06′20″W	300	Tower		12.0					
TS-6	5/25/52	11:59:59.6	37°05′44″N 116°06′20″W	300	Tower		11.1		0.04	0.6		
TS-7	6/1/52	11:54:59.8	37°02′53″N 116°01′16″W	300	Tower		14.6		0.13	0.8		
TS-8	6/5/52	11:55:00.3	37°08′19″N 116°07′04″W	300	Tower		13.9		0.08	0.8		
Operation Ivy												
Mike	10/31/52	19:14:59.4	11°14′14″N 162°11′47″E	20	Tower	14			0.06	1.2	1.0	
King	11/15/52	23:30:	11°33′44″N 182°21′09″E	1,480	Air drop							
Operation Upshot Knothole												
Annie	3/17/53	13:20:00.3	37°02′52″N 116°01′16″W	300	Tower		16.2		0.15	0.8		

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Nancy	3/24/53	13:10:00.01	37°05′44″N 116°06′10″W	300	Tower		24.4	0.11	0.6
Ruth	3/31/53	13:00:00.0	37°04′58″N 116°01′26″W	300	Tower		0.2	No motion	...
Dixie	4/6/53	15:29:38.4	37°05′05″N 116°01′05″W	6,020	Air drop		10.9	0.14	1.0
Ray	4/11/53	12:44:59.8	37°05′56″N 116°05′33″W	100	Tower		0.2	No motion	...
Badger	4/18/53	12:35:00.0	37°08′18″N 116°07′04″W	300	Tower		23.0	0.20	1.0
Simon	4/25/53	12:29:59.8	37°03′11″N 116°06′10″W	300	Tower		42.7	0.24	0.8
Encore	5/8/53	15:29:55.4	36°30′00″N 115°55′44″W	2,425	Air drop		26.5	0.15	0.8
Harry	5/19/53	12:04:59.5	37°02′25″N 116°01′31″W	300	Tower		32.4	0.27	1.2
Grable	5/25/53	15:30:00.3	36°47′35″N 115°54′53″W	524	Gun		14.9	0.40	0.7
Climax	6/4/53	11:14:56.7	37°05′15″N 116°01′06″W	1,334	Air drop		60.8	0.21	0.9
Operation Castle									
Bravo	2/28/54	18:45:00.0	11°41′27″N 165°16′25″E		Surface	15		0.20	1.0
Romeo	3/26/54	18:30:00.4	11°41′27″N 165°16′23″E		Barge			Record missing	...
Koon	4/6/54	18:20:00.4	11°29′48″N 165°22′03″E		Surface		About 100 KT		
Union	4/25/54	18:10:00.6	11°39′59″N 165°23′14″E		Barge			0.09	1.0
Yankee	5/4/54	18:10:00.1	11°39′56″N 165°23′13″E		Barge			0.20	1.0
Nectar	5/13/54	18:20:00.1	11°40′14″N 162°11′47″E		Barge			trace	
Operation teapot									
Wasp	2/18/55	19:59:59.2	37°05′12″N 116°01′19″W	762	Air drop		1.2	Micro-seisms	...
Moth	2/22/55	13:45:00.0	37°02′52″N 116°01′16″W	300	Tower		2.4	0.03	0.9
Tesla	3/1/55	13:30:00.3	37°07′32″N 116°02′51″W	300	Tower		6.8	Record missing	
Turk	3/7/55	13:20:00.2	37°08′18″N 116°07′03″W	500	Tower		43	0.16	0.8
Hornet	3/12/55	13:19:59.8	37°02′25″N 116°01′31″W	300	Tower		3.6	0.04	1.0

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TABLE 1. Continued

Shot Name	Date	Time, UT	Coordinates	Height of Burst, ft.	Type of Burst	Yield Mt	Yield kt	Yield Tons	P wave* Amplitude, μ	P wave* Period, sec	Rayleigh Wave† Amplitude, μ	Arrival of Air Shock at Ground
Bee	3/22/55	13:04:59.9	37°05′41″N 116°01′26″W	500	Tower		8.1		0.06	0.8		
Ess	3/23/55	20:30:00.0	37°10′06″N 116°02′38″W	−67	Underground		1.1		0.03	0.7		
Apple I	3/29/55	12:55:00.1	37°05′44″N 116°06′10″W	500	Tower		14.2		0.06	0.7		
Wasp′	3/29/55	17:59:54.8	37°05′12″N 116°03′28″W	740	Air drop		3.2		0.01	0.7		
Ha	4/6/55	18:00:04.1	37°01′43″N 116°03′28″W	36,620	Missile		3.3		No motion			
Post	4/9/55	12:30:00.2	37°07′20″N 116°02′04″W	300	Tower		1.5		Trace			
Met	4/15/55	19:15:00.3	36°47′53″N 115°55′44″W	400	Tower		22.0		0.54	0.7		
Apple II	5/5/55	12:10:00.0	36°03′11″N 116°06′09″W	500	Tower		28.5	0.21		0.9		
Zucchini	5/15/55	11:59:59.9	37°05′41″N 116°01′26″W	500	Tower		28.2					
Operation Wigwam												
Wigwam	5/14/55	20:00:00.0	28°44′N 126°16′W	−2,000	Underwater		30.0		0.4	1.0		
Operation Redwing												
Lacrosse	5/4/56	18:25:29.9	11°33′28″N 162°21′18″E		Surface		Kiloton range					
Cherokee	5/20/56	17:50:38.7	11°40′06″N 165°23′39″E	4,320	Air drop	Mt range						
Zuni	5/27/56	17:56:00.3	11°29′48″N 165°22′09″E		Surface				0.07	1.0		
Erie	5/30/56	18:15:29.3	11°32′40″N 162°21′52″E	300	Tower							
Seminole	6/6/56	00:55:30.0	11°40′35″N 162°13′02″E		Surface							
Flathead	6/11/56	18:26:00.1	11°36′00″N 165°27′05″E		Barge							

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Blackfoot	6/11/56	18:26:00.3	11°33′04″N 162°21′31″E	200	Tower					
Osage	6/16/56	01:13:53.1	11°32′48″N 162°21′39″E	680	Air drop					
Dakota	6/25/56	18:06:00.2	11°36′10″N 165°27′05″E		Barge					
Apache	7/8/56	18:06:00.2	11°40′17″N 162°12′01″E		Barge			0.04	1.0	
Navajo	7/10/56	17:56:00.3	11°39′48″N 165°23′14″E		Barge			0.15	1.0	0.35
Tewa	7/20/56	17:46:00.0	11°40′26″N 165°20′22″E		Barge			0.09	1.0	0.35
Huron	7/21/56	18:16:00.1	11°40′19″N 162°22′09″E		Barge					
Operation Plumbbob										
Boltzman	5/28/57	11:55:00.2	37°05′41″N 116°01′25″W	500	Tower	11.5			0.10	1.0
Franklin	6/2/57	11:54:59.9	37°02′52″N 116°01′16″W	300	Tower	0.14				
Lassen	6/5/57	11:45:00.3	37°08′05″N 116°02′27″W	500	Balloon		00.47			
Wilson	6/18/57	11:45:00.3	37°08′05″N 116°02′27″W	500	Balloon	10.3			0.12	0.9
Priscilla	6/24/57	13:30:00.1	37°47′53″N 115°55′44″W	700	Balloon	36.6			0.09	0.8
Hood	7/5/57	11:40:00.1	37°08′05″N 116°02′27″W	1,500	Balloon	74.3			0.42	1.0
Diablo	7/15/57	11:30:00.1	37°09′01″N 116°06′31″W	500	Tower	17.0		0.01	0.5	
John	7/19/57	14:00:04.6	37°09′38″N 116°03′11″W	30,000	Rocket	~2 kt		Record missing		
Kepler	7/24/57	11:49:59.9	37°05′— N 116°06′— W	500	Tower	10.3		0.06	0.8	
Owens	7/25/57	13:29:59.7	37°08′05″N 116°02′27″W	500	Balloon	9.7		0.11	1.0	
Stokes	8/7/57	12:25:00.2	37°05′12″N 116°01′25″W	1,500	Balloon	19.1		0.08	1.0	
Saturn	8/10/57	00:59:55.1	37°11′38″N 116°02′00″W		Underground	None				
Shasta	8/18/57	12:00:00.0	37°07′41″N 116°06′23″W	500	Tower	16.5		0.07	0.8	
Doppler	8/23/57	12:30:00.1	37°05′12″N 116°01′25″W	1,500	Balloon	10.7		0.06	1.0	

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TABLE 1. Continued

Shot Name	Date	Time, UT	Coordinates	Height of Burst, ft.	Type of Burst	Yield Mt	Yield kt	Yield Tons	P wave* Amplitude, μ	P wave* Period, sec	Rayleigh Wave† Amplitude, μ	Arrival of Air Shock at Ground
Franklin'	8/30/57	12:39:59.9	37°05′12″N 116°01′25″W	750	Balloon		4.7					
Smoky	8/31/57	12:30:00.0	37°11′14″N 116°04′04″W	700	Tower		44		0.08	0.7		
Galileo	9/2/57	12:40:00.0	37°03′11″N 116°06′09″W	500	Tower		11.4		No motion			
Wheeler	9/6/57	12:45:00.0	37°08′05″N 116°02′27″W	500	Balloon			197	No motion			
Coulomb B	9/6/57	20:05:00.6			Surface		0.3					
LaPlace	9/8/57	12:59:59.8	37°05′12″N 116°01′25″W	750	Balloon				No motion			
Fizeau	9/14/57	16:44:59.8	37°02′01″N 116°01′53″W	500	Tower		11.1		0.10	0.9		
Newton	9/16/57	12:49:59.9	37°05′12″N 116°01′25″W	1,500	Balloon		12		0.04	0.8		
Rainier	9/19/57	16:59:59.5	37°11′45″N 116°12′11″W	−790	Underground		1.7		0.18	1.0		
Whitney	9/23/57	12:29:59.8	37°08′18″N 116°07′03″W	500	Tower		18.5		0.05	0.7		
Charleston	9/28/57	12:59:59.9	37°08′05″N 116°02′27″W	1,500	Balloon		11.5		0.06	0.9		
Morgan	10/7/57	13:00:00.1	37°08′05″N 116°02′27″W	500	Balloon		8		No motion			
Operation Hardtack, Phase I												
Yucca	4/28/58	02:40:00.3	12°37′00″N 163°01′30″E	86,000	Balloon				No motion			
Cactus	5/5/58	18:15:00.1	11°33′23″N 162°21′15″E		Surface				No motion			
Fir	5/11/58	17:50:00.1	11°41′27″N 165°16′25″E		Barge				0.03	1.0	0.06	
Butternut	5/11/58	18:15:00.1	11°32′28″N 162°21′02″E		Barge				No motion			
Koa	5/12/58	18:30:00.1	11°40′30″N 162°12′20″E		Surface				0.07	1.3	0.10	

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Wahoo	5/16/58	01:30:00.5	11°20′41″N 162°10′44″E	−500	Underwater	0.06	1.2	Micro-seisms
Holly	5/20/58	18:30:00.1	11°32′38″N 162°21′22″E		Barge	No motion		
Nutmeg	5/21/58	21:20:00.2	11°29′46″N 165°22′15″E		Barge	No motion		
Yellow-wood	5/26/58	02:00:00.1	11°39′37″N 162°13′31″E		Barge	No motion		
Magnolia	5/26/58	18:00:00.1	11°32′34″N 162°21′14″E		Barge	No motion		
Tobacco	5/30/58	02:15:00.2	11°39′48″N 162°13′48″E		Barge	No motion		
Sycamore	5/31/58	03:00.00.1	11°41′27″N 165°16′25″E		Barge	No motion		
Rose	6/2/58	18:45:00.1	11°32′28″N 162°21′06″E		Barge			
Umbrella	6/8/58	18:15:00.2	11°22′51″N 162°13′09″E	−150	Underwater	0.6	1.2	
Maple	6/10/58	17:30:00.1	11°41′14″N 165°24′54″E		Barge	No motion		
Aspen	6/14/58	17:30:00.1	11°41′27″N 165°16′24″E		Barge	No motion		
Walnut	6/14/58	18:30:00.1	11°39′37″N 162°13′31″E		Barge	0.04	1.3	
Linden	6/18/58	03:00:00.1	11°32′39″N 162°21′23″E		Barge			
Redwood	6/27/58	17:30:00.1	11°41′14″N 165°24′54″E		Barge	No motion		
Elder	6/27/58	18:30:00.1	11°39′48″N 162°13′48″E		Barge	No motion		
Oak	6/28/58	19:30:00.1	11°36′28″N 162°06′28″E		Barge	0.1	1.2	
Hickory	6/29/58	00:00:00.1	11°29′46″N 165°22′15″E		Barge			
Sequoia	7/1/58	18:30:00.1	11°32′39″N 162°21′23″E		Barge			
Cedar	7/2/58	17:30:00.1	11°41′27″N 165°16′25″E		Barge	No motion		
Dogwood	0/5/58	18:30:00.2	11°39′48″N 162°13′48″E		Barge	No motion		
Poplar	0/12/58	03:30:00.1	11°41′17″N 165°15′52″E		Barge	0.1	1.1	
Juniper	7/22/58	04:20:00.1	11°29′46″N 165°22′15″E		Barge			

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TABLE 1. Continued

Shot Name	Date	Time, UT	Coordinates	Height of Burst, ft.	Type of Burst	Yield Mt	Yield kt	Yield Tons	P wave* Amplitude, μ	P wave* Period, sec	Rayleigh Wave† Amplitude, μ	Arrival of Air Shock at Ground
Olive	7/22/58	20:30:00.2	11°39′48″N 162°13′48″E		Barge				No motion			
Pine	7/26/58	20:30:00.2	11°39′22″N 162°13′11″E		Barge				0.8	1.3		
Teak	8/1/58	10:50:05.6	16°44′38″N 169°32′00″W	252,000	Rocket						0.05	10:53:16
Orange	8/12/58	10:30:08.6	16°21′30″N 169°32′08″W	141,000	Rocket						0.05	10:32:07
Operation Hardtack, Phase II												
Otero	9/12/58	20:00:00.2	37°02′60″N 116°01′55″W	−500	Underground			38	No motion			
Bernalillo	9/17/58	19:30:00.2	37°02′58″N 116°01′59″W	−500	Underground			15	No motion			
Eddy	9/19/58	14:00:00.2	37°05′12″N 116°01′25″W	500	Balloon			83	No motion			
Luna	9/21/58μ	19:00:00.2	37°02′57″N 116°02′01″W	−500	Underground			1.5				
Valencia	9/26/58	20:00:00.2	37°02′59″N 116°01′47″W	−500	Underground			2	No motion			
Mars	9/28/58	00:00:00.2	37°11′35″N 116°12′02″W		Underground			13.5	No motion			
Mora	9/29/58	14:05:00.1	37°05′12″N 116°01′25″W	1,500	Balloon		2		Trace			14:05:01.4
Hidalgo	10/5/58	14:10:00.1	37°05′12″N 116°01′25″W	377	Balloon			77				
Colfax	10/5/58	16:15:00.2	37°02′56″N 116°02′03″W	−500	Underground			5.5				
Tamalpais	10/8/58	22:00:00.1	37°11′43″N 116°12′01″W	−330	Underground			72	No motion			
Quay	10/10/58	14:30:00.1	37°05′41″N 116°01′25″W	100	Tower			79				14:30:00.1
Lea	10/13/58	13:20:00.1	37°05′12″N 116°01′25″W	1,500	Balloon		1.4		Trace			13:20:01.4
Neptune	10/14/58	18:00:00.2	37°11′38″N 116°11′59″W	−98.5	Underground			115	No motion			

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Hamilton	10/15/58	16:00:00.2	36°48′08″N 115°55′56″W	50	Tower		1.17			
Logan	10/16/58	06:00:00.1	37°11′03″N 116°12′04″W	−830	Underground	5.0		0.4	0.7	
Dona Ana	10/16/58	14:20:00.1	37°05′12″N 116°01′25″W	500	Balloon		37			14:20:00.5
Vesta	10/18/58	23:00:00.2	37°07′21″N 116°02′05″W		Surface		24			
Rio Arriba	10/18/58	14:25:00.1	37°02′28″N 116°01′33″W	70	Tower		90			14:25:00.1
Socorro	10/22/58	13:30:00.2	37°05′12″N 116°01′25″W	1,500	Balloon	6		0.02	1.0	13:30:01.0
Wrangell	10/22/58	16:50:00.1	36°47′53″N 115°55′44″W	1,500	Balloon		115			16:50:01.2
Rushmore	10/22/58	23:40:00.1	37°08′05″N 116°02′27″W	500	Balloon		188			23:40:00.3
Catron	10/24/58	15:00:00.2	37°02′35″N 162°01′37″W	72.5	Tower		21			
Juno	10/24/58	16:01:00.2	37°07′24″N 116°02′16″W		Surface		1.7			
Ceres	10/26/58	04:00:00.2	37°10′53″N 116°04′07″W	25	Tower		0.67			
Sanford	10/26/58	10:20:00.1	36°47′53″N 115°55′44″W	1,500	Balloon	4.9		0.07	0.8	10:20:00.8
De Baca	10/26/58	16:00:00.1	37°05′12″N 116°01′25″W	1,500	Balloon	2.2		Trace		16:00:01.3
Chavez	10/20/58	14:30:00.2	37°02′41″N 116°01′47″W	52.5	Tower					
Evans	10/29/58	00:00:00.2	37°11′41″N 116°12′17″W	−848	Underground		55	No motion		
Humboldt	10/29/58	14:45:00.1	37°02′52″N 116°01′29″W	25	Tower		7.8	No motion		
Santa Fe	10/30/58	03:00:00.1	37°05′12″N 116°01′25″W	1,500	Balloon	1.25				03:00:01.5
Blanca	10/30/58	15:00:00.2	37°11′09″N 116°12′07″W	−835	Underground	19.2		0.7	0.7	
Titania	10/30/58	20:34:00.2	37°10′38″N 116°04′09″W	25	Tower	0.15				

* *P* wave amplitudes and periods at Pasadena from Benioff short-period vertical.
† Rayleigh wave amplitudes for period of 20 seconds from Benioff long-period vertical and Press-Ewing long-period vertical.

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TABLE 2. Energy Coupled to Seismic Waves and Rough Equivalent P Wave Magnitude for a 20-kt Nuclear Explosion under Various Conditions

	E_s^*/E_0	Magnitude
10 km air burst	1×10^{-6}	3
1 km air burst	3×10^{-5}	3½
Surface	1×10^{-4}	4
300 m underground	1×10^{-3}	5
30 m underwater	5×10^{-3}	5½
100 m underwater	2×10^{-2}	6
500 m underwater	4×10^{-2}	6½

* Ratio of seismic energy to total energy (yield).

deep 25-kt underwater explosion is equivalent to a 10-Mt surface explosion. Thus, for seismic research, it appears that deep underwater explosions are most desirable, with underground explosions a poor second. If the underwater tests have a scale depth of 500 meters for 1 kt (5 km for 1 Mt), there will be very little surface disturbances, since the gas bubble will reach the surface only after about 8 pulsations, by which time it will have been reduced to negligible size.

The largest nuclear explosion which could be safely contained in the ocean is about 10 Mt. This would produce distant P wave amplitudes corresponding to those of the largest earthquake man has ever recorded, and may be larger than is needed. Such a large explosion would best be set off in one of the deepest of the ocean trenches. These trenches are all seismically active, so one would lose the advantage of producing a seismic source in an aseismic area.

A clean 1-Mt explosion could safely be conducted almost anywhere in the deep ocean. Very little, if any, radioactivity would be brought to the surface if the explosion were at a depth of 5 km. The radioactivity would be mixed with an enormous volume of water and would be diluted so quickly that no contamination hazard to commercial seafood could occur. The total activity would be comparable to that in the Wigwam test, and it would be diluted very much more before it rose to the shallow depth of commercial fishing. No contamination of fish was found at Wigwam. The total fission product radioactivity would be comparable to that produced in 1 day by a Hanford reactor. There would be no damage to shipping outside the immediate area of the explosion, although the explosion would be felt at great distances.

A deep underwater 1-Mt explosion would give P wave amplitudes corresponding to a magnitude-8 earthquake. Earthquakes of this size occur only about once every 3 years, always in the small regions of high seismicity. Since one-quarter of the earth's surface is covered by oceans more than 5 km deep, 1-Mt shots could be conducted in a wide variety of regions where large earthquakes have never occurred.

One-Mt explosions would be completely contained on land, if detonated at a depth of 1.5 km (5000 ft). It is difficult to compare a megaton explosion underground with a corresponding underwater burst because we lack experimental data or appropriate theory. However, such explosions would be recorded world wide.

It is perhaps worthy of mention that several sources, appropriately disposed, may be detonated in a time sequence to produce directional effects, to favor the propagation of a particular wave mode, or, possibly, to excite the fundamental modes of oscillation of the earth as a whole.

Since we are considering explosions in general, we should consider equally the possibilities of chemical explosions. Whenever a chemical explosive will do the job as well and the cost is not excessive, it is clearly preferable to a nuclear explosive. Chemical explosives have not been exploited enough in terms of large explosions planned and detonated for seismic purposes. Soviet scientists have shown how much detailed information on crustal structure can be obtained with chemical explosives.

The cheapest of chemical explosives costs about ten cents a pound. Figure 2 shows the ratio of cost of chemical explosives at this figure to the cost of nuclear explosives released by the AEC [*Vortmann*, 1958, p. 70]. Since the cost of a nuclear explosive does not vary rapidly with yield, the ratio of costs is nearly linear. This suggests that all explosions up to a few kilotons should be chemical, unless the cost of emplacing them is large compared with the cost of the explosive.

Whereas it is highly desirable that nuclear explosions on land be completely contained, to prevent dissemination of the radioactivity, chemical explosions do not need to be contained.

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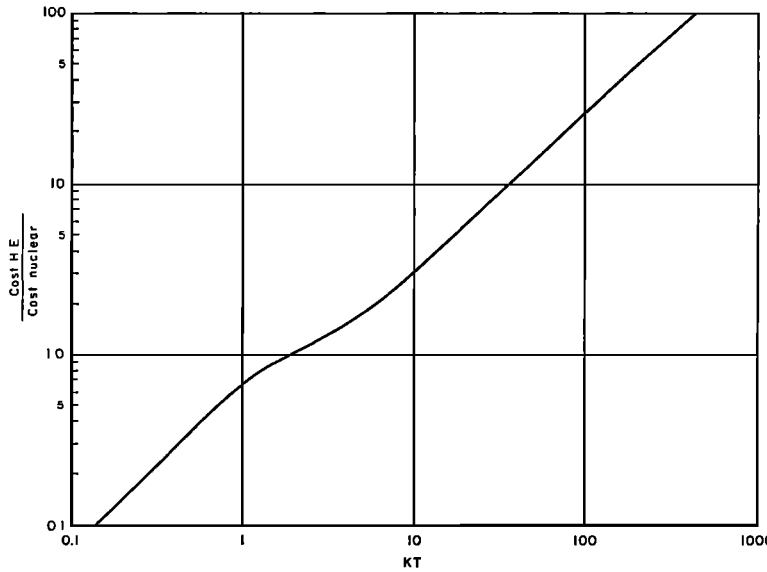

Fig. 2. Relative cost of high explosive and nuclear devices versus yield.

The most effective way to use large chemical explosions on land for seismic purposes is to detonate them in deep lakes. A kiloton of chemical explosive fired at a depth of 100 meters in a lake 200 meters deep should give a stronger seismic signal than the 20-kt Blanca shot, which was detected by standard seismic stations in Europe.

So far, we have talked entirely about the sources, but the greatest rewards will come only from much greater emphasis on instrumentation, and on the theory of seismic sources and propagation. Then, once precise seismic data are in hand, we will need advances in solid-state theory and experiment to derive from these seismic data the maximum information as to the constitution and temperature of the interior of the earth.

Proposed International Program of Explosions for Seismological Research

The program we propose is based on the assumption that funds will be available for seismology commensurate with its importance to the science of geophysics. Although nuclear explosions form an essential part of this program, the scientific potential of such explosions cannot be realized without great augmentation and improvement of seismic instrumentation. Each nuclear explosion should also be preceded by a series of chemical explosions to set the stage for reaping the full rewards from these unique sources of seismic energy.

Goals of the proposed program

We shall first list the scientific objectives of the proposed program and then outline a way in which these objectives might be achieved. The goals are as follows:

Rounding out the seismic coverage of the earth. The aseismic regions of the earth are little known seismically. For example, since Antarctica has very few earthquakes, little is known about its structure. Only meager information [e.g., *Press and Dewart*, 1959] is available as to whether it possesses typical continental structure, whether it has a predominantly oceanic crust, or whether it is intermediate in character. Surface waves excited by a large nuclear explosion in Antarctica or in the deep ocean near Antarctica, if recorded by appropriately located stations in Antarctica, could be used to determine the thickness and extent of the continental crust. Higher-mode surface waves from a buried nuclear explosion within Antarctica may yield the over-all average ice thickness, data much sought after by geophysicists.[3]

[3] Since this was written an Antarctic Treaty has been presented which precludes nuclear explosions in Antarctica.

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The seismic coverage of the earth is also spotty because of the distribution of seismic observatories. The system of seismic stations proposed by the Geneva Conference of Experts would vastly improve world-wide seismic coverage and could greatly augment the seismic results to be obtained by the program outlined below.

Refinement of velocity versus depth relations and delineation of the differences between mantle velocities under continents and under oceans. To this end, it is necessary to obtain much more accurate travel-time curves and amplitude versus distance relations through comprehensive refraction profiles with closely spaced seismometers and accurate timing. Surface-wave dispersion studies must be correlated with these refraction profiles, since surface waves give valuable additional information as to the average elastic properties of the layers in which they are propagated. Reflection studies should be made in the degree to which they are possible.

These efforts must be planned so as to reveal the details of the low-velocity layer (or decreased-velocity gradient) in the outer few hundred kilometers of the mantle. Of special interest are regional differences in the behavior of this layer, e.g., under continents and oceans. Oddly enough, there is more argument among seismologists about the seismic character of this outer shell of the earth than about any of the deeper parts of the earth. This is precisely the region which it is essential to know well in order to arrive at valid theories of the origin of continents and oceans, the genesis of mountains, and the source of volcanic lavas, ore bodies, diamond pipes, and all other geologic features which have their origin below the continent crust.

Increased knowledge of the earth's core. The presumption that the earth's core is composed of an outer shell of molten iron and an inner core of solid iron rests on shaky evidence—the fact that shear waves are not propagated through the outer core, for example, and the analogy between the mantle and the core on the one hand and the stony and iron meteorites on the other. It is possible to learn very much more about the core with nuclear explosions and improved instrumentation.

The outer boundary of the core has been shown by earthquake seismology to be a transitional region some 200 km thick. It is highly desirable to refine these data by explosions in selected locations with closely spaced seismometers deployed on a line at the appropriate distance.

With a large underwater explosion, efforts should be made to detect multiple reflections from the core boundary to get additional information on the velocity gradient and possibly the density contrast. Attempts should also be made to learn the details of S-wave attenuation in the transitional zone.

If the inner core is solid, it should transmit shear waves formed by P-S conversion at its boundary. A search should be made for the $PKJKP$ wave whose travel time and amplitude have been predicted by *Bullen* [1951].

The accuracy of velocity determinations within the core is limited by inadequate knowledge of time delays due to crustal structure in the neighborhood of seismic stations. Two remedies are suggested: (1) Determine the local seismic velocities on the appropriate arrival paths by a series of refraction and reflection profiles (2) Deploy seismometers on the deep ocean floor where the crust is known to be thin and uniform. Either of these is relatively expensive, but both would be very rewarding in providing all kinds of by-product information.

Resolution of other outstanding problems of seismology (typical examples). (*a*) Long-period G waves ($T \sim$ 100–200 sec) from large earthquakes are received at Pasadena from both directions around the earth, and hence the distance to the epicenter can be determined. Epicenters determined in this manner occasionally differ by as much as 200 km from the epicenters determined for the same earthquake by the short-period P waves. Also amplitudes show a corresponding sensitivity to the direction in which the waves left the source. Large underwater nuclear explosions might generate long-period surface waves, allowing a similar comparison. If such a difference in apparent epicentral position is found for explosion sources, the effect is one of propagation. If these differences are not found in waves from explosions, the effect is one of earthquake mechanism and may be used to learn more about the nature of earthquakes, especially the extent of faulting.

(*b*) *Ewing and Press* [1956] suggested that the long-period nature of S waves is due to preferential attentuation in the outer mantle.

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They further suggested that this attenuation is due to local pockets of molten or partially molten material in the mantle. This may reflect a high-temperature zone in the outer mantle at a depth of 100 to 300 km and may be related [*Press*, 1959] to the low-velocity layer for which evidence has been accumulating. There is some evidence that regional differences exist for this effect.

This attenuation can be investigated in detail by long refraction profiles using nuclear and large chemical explosions in the appropriate regions, provided adequate shear waves are generated.

(*c*) A search is in order for seismic differences in the outer mantle related to the large differences in surface heat flow observed in the oceans. Heat flow through the Pacific floor varies by more than a factor of 50 from one place to another. Such variation can hardly be due to conditions in the crust, but must come from different temperature gradients in the mantle. Seismic studies from explosions have not yet penetrated any appreciable distance below the Mohorovicic discontinuity anywhere and may not have penetrated it at all in the region of the highest observed heat flow—the Albatross plateau off the west coast of South America.

Large chemical explosives probably will suffice to plumb the depths sufficiently to resolve this question.

(*d*) A similar search should be made for the mechanism of isostatic compensation of oceanic regions of different elevation that are in isostatic balance. For example, on opposite sides of the Mendocino escarpment, the elevation is different by more than a kilometer, yet the acceleration of gravity at sea level is the same on both sides. The difference in density necessary to maintain this balance should be detectable seismically, but it has not been found in the shallow refraction studies to date.

Determination of the nature of earthquakes. Since explosions are implusive sources of short duration, the seismometer output is the response of the earth for transmission of the known source impulse. Hence it is possible, in principle, once this response is known, to work backward from earthquake records to determine the true nature of earthquake sources. This would have great value for tectonic studies. It is also pertinent to the problem of discriminating between earthquakes and explosions in the event of a test moratorium.

Specific Proposals

Many possible experiments will occur to geophysicists. We outline below one possible program to illustrate the magnitude of the effort we have in mind. The final program must be decided by an international group of scientists.

Exploitation of Plowshare detonations. Each explosion planned in the Plowshare program constitutes a unique source of seismic waves. It is to be hoped that appropriate steps will be taken to insure that the maximum seismological benefit will be derived from these explosions. To this end, it is necessary that the detonation time and position be published long enough in advance to allow the various seismological organizations to plan and deploy their instruments. It would be desirable if, in each Plowshare project, a series of chemical explosions could be detonated before the nuclear explosion, to provide calibration of seismometers.

Plowshare explosions such as the shot in salt (Gnome), the proposed test shot in the Athabaska tar sands, and the proposed harbor excavation project (Chariot) would constitute seismic sources of unusual interest, since they are all in regions of interesting geology where there have been no large explosions for seismic purposes. All the other proposed Plowshare projects would also be very valuable seismic sources.

In particular, each Plowshare explosion should be used as a source for several refraction profiles radiating from it. We propose that several hundred automatic or remote-controlled seismic recorders be deployed for this purpose at each Plowshare explosion. An effective refraction line to probe the outer few hundred kilometers of the mantle should be composed of about a hundred sensitive seismometer arrays on a line 2500 to 3000 km long. When several such lines are established, radiating from the position of the nuclear explosion, a series of large chemical explosions should be conducted along each line to determine local structure and thus reap the maximum benefit from this extensive seismic array. The unmanned seismic stations proposed by the *Panel on Seismic Improvement* [1959] would be admirably suited for deployment on the Plowshare lines. Such a use, prior to installing them in the Geneva network for seismic detec-

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tion of A-bomb tests, would constitute an excellent field test for the equipment and would provide much-needed information on the nature of seismic signals from a nuclear explosion in addition to the purely scientific rewards.

A series of such profiles associated with Plowshare projects like Gnome, Chariot, and the Athabaska tar sand test shot would provide seismology with far more detailed and accurate information than has been obtained from earthquakes and chance quarry blasts in fifty years. We would begin to know in detail what the underpinnings of continents are like.

These profiles should be augmented by other seismic detectors—notably a coarser net of long-period instruments to utilize the surface waves emanating from the Plowshare shots.

Oceanic seismometer lines. In many respects the ocean floor is the best site for seismological stations. The multitude of short (100–150 km) refraction profiles that have been made by Ewing, Press, Officer, Raitt, Shor, and others have shown that the crust beneath the deep ocean floor is thin and remarkably uniform. The evidence at hand indicates that seismic background noise is low. Hence, high-sensitivity seismometers with precision timing can provide far more accurate information on the characteristics of the mantle than may ever be possible from typical continental sites. The 'station error' which plagues ordinary seismological observations results from local variations in velocity. In continental stations, seismic waves coming from different directions will have travel times in the last hundred kilometers different by as much as several seconds, averaging about ½ second for 'good' observatories.

The deep oceanic crust is about one-eighth as thick as the continental crust, and its variation in velocity is about one-third that in the continental crust. Hence the inherent error for an uncalibrated station should be about 0.02 seconds, and this may be reduced by calibrating the stations with local refraction profiles. Whereas conventional seismological stations report arrival times to the nearest second, deep ocean stations should report to the nearest tenth or possibly hundredth of a second.

We propose that several lines of seismometers be established on the floor of the ocean. Each line should be 2500 km long and should be composed of 100 evenly spaced stations. Ninety of these could be arrays of ordinary hydrophones. Every tenth station should also contain three-component short- and long-period seismometers placed on the bottom. There are various ways in which this can be done—all expensive. Perhaps the best way, considering all aspects of the problem, is to lay a single long cable with the seismometers attached to it, terminating on an island or continental shore. The seismometers would receive power through the cable, and would transmit their signals to the land station for recording and processing. Alternative methods of establishing such a line are to use telemetering buoys or recoverable ocean-bottom instruments.[4]

The position of each seismometer must be known to a high degree of accuracy if the full potential of these seismometer lines is to be realized. Long-range travel times were determined to an accuracy of about 0.2 per cent by earthquake seismology. We are striving to make a substantial improvement in this accuracy. The most promising method of location seems to be to determine the travel times of the direct water-borne shock wave from explosive sources to the seismometers. Deep water velocity is now known to about 0.1 per cent. This accuracy must and can be improved. Two methods are obvious: (1) Collect samples of bottom water and measure the velocity in the laboratory at the appropriate pressures and temperatures. Such velocity measurements can be made to accuracies of better than 0.01 per cent. (2) Determine the distance between the stations by radio ranging methods and thus calibrate the average water velocity, correcting locally for changes due to temperature, pressure, and salinity.

Figure 3 shows a schematic layout of an oceanic seismometer line together with a suggested schedule of 10-, 100-, and 1000-ton chemical explosive shots. The 10-ton shots can be recorded to distances of about 300 km and will penetrate the mantle to a depth of about 80 km. The 100-ton shots can be recorded to 1000 km or

[4] A recent experiment by the Lamont Geological Observatory, with a self-contained bottom seismometer, demonstrated that seismic noise on the sea floor is exceedingly small—so much so that bottom seismometers seem much the best technique for conducting this experiment. This remarkable advance might make possible a significant decrease in shot size over those given below.

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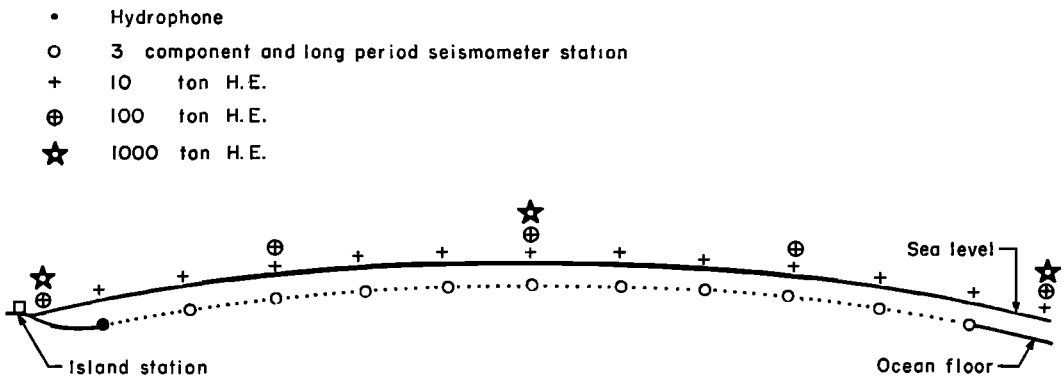

Fig. 3. Deep ocean seismometer line and local shot schedule.

more and should provide information about the low-velocity layer in the upper mantle. The 1000-ton shots can be recorded to at least 3000 km and will penetrate to perhaps 700 km. We suggest that there be thirteen 10-ton shots, one at each 250-km interval and one 250 km each way past the end of the line. There should be five 100-ton shots, spaced at 750-km intervals as indicated in Figure 3. Five 1000-ton shots are also needed, three at 1300-km intervals and two more 1300 km beyond each end of the line.

This program would probe the outer critical 700 km of the mantle over an arc of 30° with undreamed-of accuracy. Not only would it be the first seismic exploration of any oceanic sector but the precision with which the velocity could be measured might make possible a correlation with gravity anomalies and small temperature changes. Thus, critical evidence for the presence of subcrustal hot spots, possible convection current cells, differences in intrinsic density, phase changes, etc., will be provided.

In addition to its utility for this local seismic survey, the oceanic seismometer line will provide an unparalleled means for observing the nature of the earth's deep interior, if nuclear explosive sources are used at larger distances. Figure 4 shows a typical array of underwater nuclear shots that would exploit the capabilities of this array of detectors. We propose six 20-kt and three 1-Mt nuclear explosions disposed as indicated. In addition, we propose that two other oceanic seismometer lines be established as shown in Figure 4.

The 1-Mt explosion at *A* will provide seismometer lines I and II with waves that have traversed the inner core, and it will provide a crucial test for the solid versus fluid hypotheses of the inner core. Both lines I and II cover the critical are of 130° and 140° from the source, which Bullen has shown is the most favorable region for detecting the ray *PKJKP*, the seismic wave that will propagate through the inner core as a shear wave if the inner core is solid. In addition, this shot is favorably positioned to produce multiple-reflected waves from the boundary of the outer core at line III.

The 1-Mt explosion at *B* will provide lines I and II with waves reflected from the outer core and possibly from the inner core. It will provide line III with diffracted waves from the outer core, and possibly with *PKJKP*. In addition, first arrivals from the shot at *B* received on lines I and II will give the distribution of velocities in the mantle at depths from 500 to 1500 km; while shot *A* will have given the values for 1500 to 2500 km at line III. The 1-Mt shot at *C* will provide line I waves diffracted from the outer core, and lines II and III waves reflected from the outer core. This shot will also give velocities from 500- to 1500-km depth in the zones of lines II and III.

A typical set of locations for the three seismometer lines would be as follows:

Line I: Straddling the Azores on a bearing of 120° to 300°, from Madeira across the abyssal plain to Horta, extending across the western slope of the mid-Atlantic ridge to about 43°N, 39°W.

Line II: NE from Hawaii to about 35°N, 137°W, traversing a well-studied region of the deep Pacific and crossing the Murray fracture zone at 32°N, 142°W.

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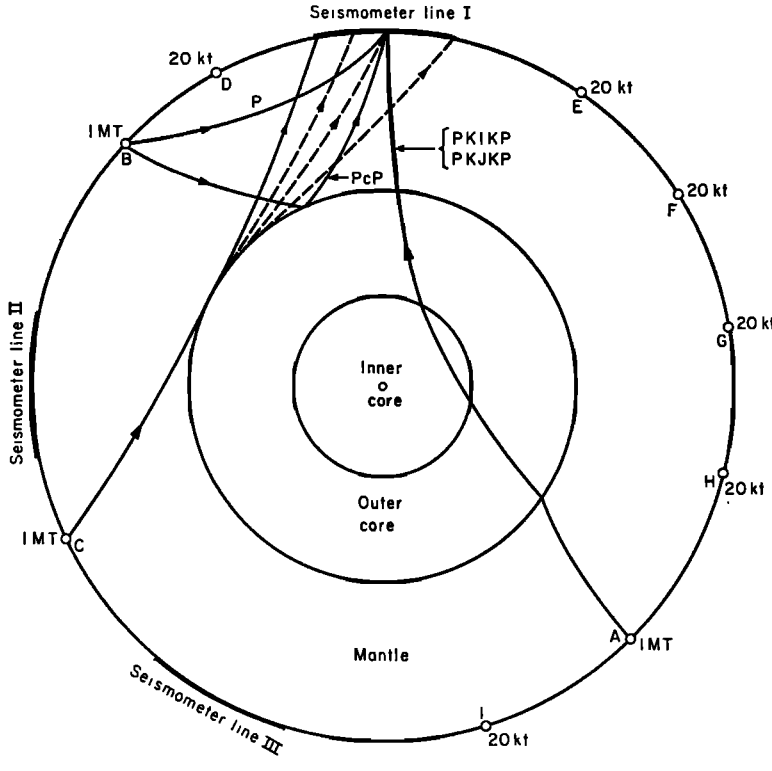

Fig. 4. Schedule of underwater nuclear explosions for oceanic seismometer lines.

Line III: NNE from North Cape, New Zealand, to Niue (Savage) Island at 19°S, 170°W, traversing the South Fiji basin, the Fiji-Kermadec rise, and the Tonga trench near its deepest point.

For this disposition of seismometer lines, the three 1-Mt explosions could be: (1) off Antarctica in the southern Indian Ocean, at about 60°S, 90°E; (2) off Africa in the depths of the Gulf of Guinea, at about 5°S, 5°E; (3) in the deep Pacific east of the Marshall Islands. There could be one 20-kt shot in the Atlantic at about 50°N, 40°E; one at the location of Wigwam, 29°N, 126°W; one south of Johnston Island at 5°N, 165°W; one south of Australia at 60°S, 150°E; and one off South Africa at about 40°S, 5°E.

These shots would all be recorded world wide, so that the information from the oceanic seismometer lines would be supplemented by a wealth of information from the other stations. If the Geneva net were installed, the records of these explosions would be obtained at these high-quality stations.

For detailed knowledge of the earth's mantle and core, however, it is anticipated that the oceanic seismometer lines will be far superior because of the greater uniformity of the material in the vicinity of the stations. We believe that the information obtained from these oceanic lines may resolve the following major geophysical questions:

(*a*) Is the inner core liquid or solid?

(*b*) What is the nature of the outer boundary of the core?

(*c*) What is the difference in the mantle under continents and oceans?

(*d*) Are there variations from place to place in the mantle under the oceans, as would be required for convection currents in the mantle?

(*e*) What is the nature of the '20° discontinuity' in these oceanic regions?

Work in related fields will be stimulated by these results and can be expected to produce far-reaching results. For instance, the more accurate knowledge of velocity distribution in

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the earth will certainly stimulate experimental and theoretical physicists to provide better equation-of-state data and theory so that the composition and temperature of the earth's layers may be more unambiguously determined.

In the process of establishing and shooting these lines, all related geophysical measurements should be made; gravity, magnetic, and heat-flow surveys and connected crustal seismic sections by means of conventional ship-to-ship shooting. From these and the mantle seismic data it will be possible to delineate the features of the oceanic crust and the outer mantle in great detail over these interesting regions where only fragmentary data have been obtained in the past.

It will be noted that these two programs—the exploitation of the Plowshare explosions and the oceanic seismometer lines with their explosion schedule—have largely covered the need for data to attain the goals described above. Other, smaller programs will be needed to fill in the gaps. These are as follows:

Programs to resolve other outstanding problems.

(*a*) A few more 20-kt underwater shots around Antarctica and other aseismic regions to provide knowledge of the crust unattainable by earthquake means. These must be done in conjunction with extensive land instrumentation and a program of large chemical explosions on land to provide local data.

(*b*) Search for the cause of attenuation of *S* waves in certain oceanic paths (local magma chambers?). The 20-kt shot in the North Atlantic as recorded on the Azores line and the high-explosive shots on that line will provide some information on this problem and may even provide the complete answer. It seems clearly desirable, however, that additional seismometers be deployed in the North Atlantic to receive the seismic waves from this 20-kt shot in other azimuths known to be more favorable. We will not go into this in detail here, since present information is not adequate to specify the best way to conduct this search. Better information will surely be available from earthquake data before plans for such an undertaking are settled.

(*c*) Search for seismic differences related to variations in oceanic heat flow and the search for the mode of compensation of oceanic regions of differing elevation. The outstanding example of these is the Albatross plateau. The necessary data may be obtained by large high-explosive shots, but may require one more 20-kt shot.

(*d*) Determination of the nature of earthquakes. This will come out of all the above programs as detailed information becomes available as to the propagation characteristics of the medium from these nuclear explosions. It may well be that the oceanic seismometer lines will provide superior information on distant earthquakes. Line III, across the Tonga trench, would be in a region of high seismicity, and would certainly provide unique information on local earthquakes if it were kept in operation long enough.

(*e*) Excitation of the free modes of oscillation of the earth. Very long period seismometers should be constructed and deployed to observe all the previously mentioned shots, as well as earthquakes. The records obtained from the 1-Mt underwater shots should be carefully analyzed in a search for evidence of excitation of the modes of oscillation of the earth—ranging from 10 to 60 minutes in period. Now that free oscillations have been observed for earthquakes, comparison with explosion-induced oscillations should reveal the properties of the earthquake source.

Conclusions

In this article, we have proposed programs for purely scientific seismic exploration of the world using high-explosive and nuclear explosions. These programs will be expensive, but the total cost will be only a small fraction of the cost of installing and operating the Geneva network of seismic stations. The scientific rewards will be tremendous and may even serve to improve the effectiveness of the Geneva network. These programs are international in scope, and they must be done on an international basis in the tradition of seismology and of the IGY. As such, we believe that they will constitute a very important aspect of the use of nuclear explosions for peaceful purposes.

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painstakingly checking the accuracy of the entries in Table 1. Dr. Jack Oliver critically read the manuscript and improved it. This work was in part carried out at the RAND Corporation under an AEC contract.

Note added in proof. A paper by *Kogan* [1960] has just come to our attention in which interesting results about the shear velocity in the upper mantle and the dimensions of the core were found by using data from nuclear explosions. The precision needed to reach these conclusions was not previously achieved from earthquakes.

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